REPORT

ON

"Retrofittable Programmable Electronic Load for Laboratory Kit Protection"

SUBMITTED

TO

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BY

Name	PRN	Roll Number	Mobile Number	Email ID
Manas Kulkarni	22311608	311058	7972470825	manas.22311608@viit.ac.in
Samiksha Nalawade	22311696	311063	8263990112	samiksha.22311696@viit.ac.in
Rajlakshmi Desai	22311732	311065	8605796792	rajlakshmi.22311732@viit.ac.in

Class: T. Y. B. Tech **Division: A** **Batch: A3**

Faculty In-Charge/Evaluator

Dr. Rahul Pol

Abstract

A Programmable Electronic Load (PEL) is an active load capable of drawing a controlled, user-defined current from a source independent of voltage variations. In this project, a PEL is designed and simulated using an AC–DC full-wave bridge rectifier followed by a buck-type DC–DC converter with closed-loop PWM control implemented using an NE555 oscillator and an LM358 dual operational amplifier.

The design satisfies all key PBL constraints:

- **Topology requirement** is met using a rectifier followed by a DC–DC converter,
- **Control technology** is implemented using PWM with feedback-based regulation,
- **User control** is achieved through smooth and continuous adjustment of load current.

The system enables adjustable current control, incorporates hardware overcurrent protection, and provides visual indication through green and red LEDs for normal and fault conditions respectively. Simulation results confirm stable PWM operation (~48 kHz), accurate current regulation, and reliable fault detection.

A key application of this design is the **protection of power electronics laboratory experiment kits**, as it prevents excessive current conditions. The system is also **retrofittable**, allowing it to replace conventional resistive loads without modifying existing setups.

1. Introduction

Modern power electronic systems require reliable testing under varying load conditions. Traditional resistive loads lack flexibility and cannot maintain constant current under changing voltages.

A Programmable Electronic Load solves this problem by:

- Maintaining **constant current**
- Allowing **user-controlled load variation**
- Providing **safe testing environment**

The proposed system uses:

- PWM-based control
- Closed-loop feedback

- Switching power electronics (buck converter)

This makes it efficient, accurate, and suitable for laboratory testing applications.

2. Related Work

Existing load testing methods include:

- **Fixed Resistive Loads**
  - Simple but not adjustable
  - Poor accuracy under varying voltage
- **Electronic Loads using Linear Regulation**
  - Accurate but inefficient (high power loss)
- **Digital/Microcontroller-based Loads**
  - Flexible but complex and costly

Limitations of existing systems:

- Lack of efficient switching control
- Poor response to dynamic conditions
- Limited protection mechanisms

Proposed system improvement:

- Uses **PWM + buck converter → high efficiency**
- Uses **analog control → simple and reliable**
- Includes **overcurrent protection + LED indication**

3. System and Circuit Model

3.1 Overall Architecture

The system consists of six major blocks:

- AC-DC Rectifier (BR1)
- Control Supply (12V & 5V)
- PWM Generator (NE555)
- Error Amplifier (LM358)
- Buck Converter (MOSFET + L (Inductor) + D (Freewheeling Diode)
- Current Sensing & Display

3.2 Functional Working

- AC input → converted to DC using bridge rectifier
- PWM signal generated using NE555 (~48 kHz)
- LM358 compares:
  - ➢ Actual current (feedback)
  - ➢ Reference signal (PWM)
- MOSFET switching controls load current
- Inductor smooths current
- RSENSE provides feedback voltage

3.3 Key Design Features

- Closed-loop current regulation
- High-frequency PWM switching
- Smooth current using inductor
- Separate supply for control and power circuits
- Real-time measurement using probes

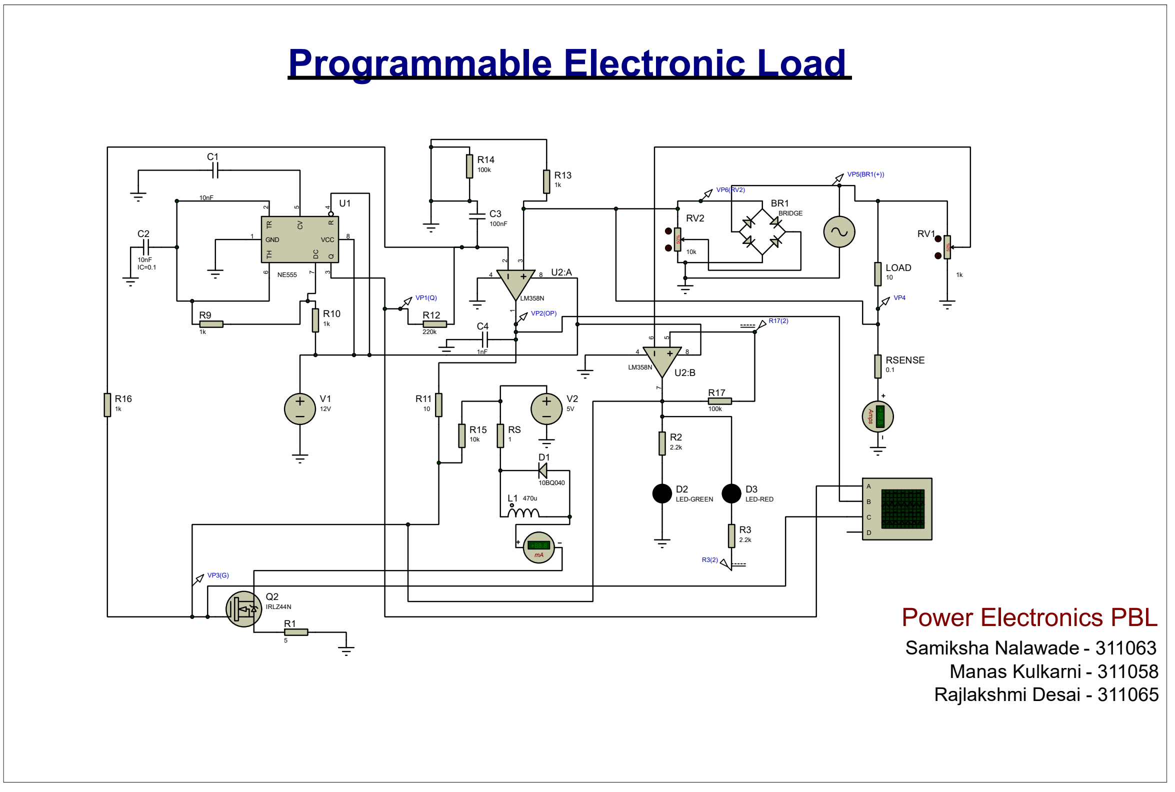

Fig.1: Circuit Simulation on Proteus

4. Circuit Description

4.1 PWM Generation (NE555)

- Operates in astable mode
- Frequency ≈ 48 kHz
- Provides control signal for regulation

4.2 Error Amplifier (LM358)

- Compares:
  - Sensed current
  - Reference PWM
- Generates control voltage for MOSFET

4.3 MOSFET Switching (IRLZ44N)

- Acts as main control element
- Logic-level MOSFET → driven directly
- Controls load current via duty cycle

4.4 Buck Converter

- Inductor stores energy
- Diode provides freewheeling path
- Ensures smooth current flow

4.5 Current Sensing

- RSENSE = 0.1Ω
- Converts current to voltage
- Used for feedback and protection

4.6 Overcurrent Protection

- Comparator checks threshold (RV2)
- Red LED → fault
- Green LED → normal operation

5. Circuit Operation

Step-by-step Operation

1. Power ON → supplies activated

2. NE555 generates PWM signal
3. LM358 compares feedback with reference
4. MOSFET switches ON/OFF at high frequency
5. Inductor smooths current
6. Current stabilizes at set value
7. If current exceeds limit → Red LED ON

6. Results and Observations

- Stable current regulation achieved
- Smooth waveform due to inductor filtering
- Fast response to load changes
- Accurate sensing via RSENSE
- Reliable overcurrent detection

Typical observations:

- PWM frequency ≈ 48 kHz
- LED indication works correctly
- Load current varies with control input

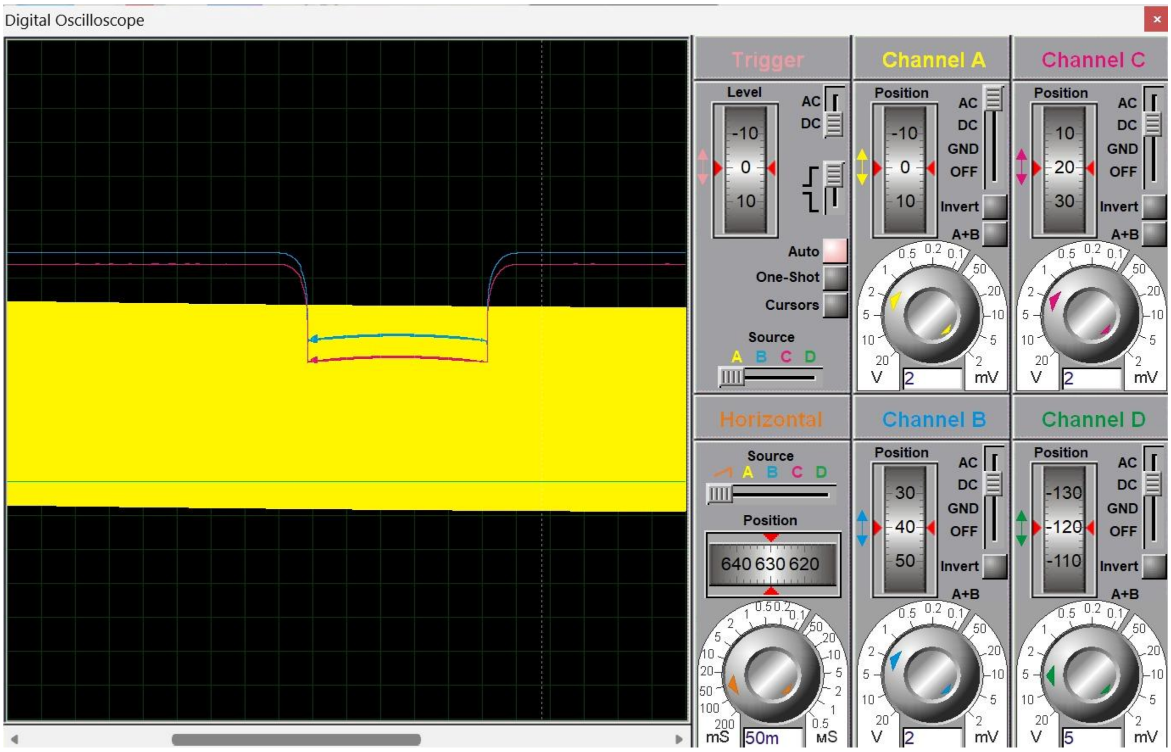

Fig.2: Digital Oscilloscope Waveform Showing PWM-Controlled Current Regulation

7. Applications

- Power supply testing
- Battery discharge testing
- Converter and inverter testing
- Laboratory experiments
- Educational demonstrations

8. Advantages

- High efficiency (switching control)
- Simple analog design
- Accurate current regulation
- Low cost and easy implementation
- Built-in protection

9. Limitations

- Limited scalability for very high power
- Analog tuning required
- No digital interface/automation
- Accuracy depends on component tolerance

9. Conclusion & Future Scope

The Programmable Electronic Load (PEL) successfully demonstrates stable closed-loop current control using PWM, MOSFET switching, and op-amp feedback. The circuit ensures reliable operation with integrated overcurrent protection and consistent current regulation.

All three PBL constraints: **topology, control technology, and user control** are effectively fulfilled. The system serves as a **practical and cost-effective solution for protecting power electronics laboratory kits**, preventing damage due to uncontrolled loading. Its **retrofittable nature** allows seamless replacement of traditional resistive loads without requiring changes to existing experimental setups, improving both safety and flexibility.

The current design can be further enhanced with the following improvements:

- **Digital Control Integration**
    - Replace NE555 with microcontroller (Arduino/STM32)
    - Enables programmable current profiles and automation
- **Display & User Interface**

  - Add LCD/OLED to show voltage, current, and power
  - Improves usability and real-time monitoring
- **Higher Power Capability**
  - Use heat sinks, better MOSFETs, and cooling systems
  - Allows handling of higher current loads
- **Multiple Operating Modes**
  - Constant Current (CC) → already implemented
  - Add:
    - Constant Voltage (CV)
    - Constant Power (CP)
    - Constant Resistance (CR)
- **Data Logging & Communication**
  - USB/Wi-Fi/Bluetooth interface
  - Useful for analysis and remote monitoring
- **Protection Enhancements**
  - Thermal protection
  - Short-circuit protection
  - Reverse polarity protection
- **PCB Design & Hardware Implementation**
  - Convert simulation to real hardware
  - Optimize layout for noise reduction and efficiency